

- 2 (a) At top of trajectory, vertical component of velocity is zero and horizontal component of velocity is constant, hence it will have minimum KE [B1]
or
At top of trajectory, gravitational potential energy of object is greatest hence kinetic energy is the lowest since only 2 forms of energies are involved. [B1]

- (b) Min KE is when object is at the highest point with only horizontal velocity.
 $\frac{1}{2} m v_x^2 = 12.5 \text{ J}$

$$v_x = u_x = \sqrt{\frac{12.5}{0.5 \times 0.25}} = 10 \text{ ms}^{-1} \quad [\text{M1}]$$

$$u \cos 40^\circ = 10 \quad [\text{M1}]$$

$$u = 13 \text{ ms}^{-1} \quad [\text{A0}]$$

(c) (i) $s_y = u_y t + \frac{1}{2} a t^2$

taking upwards as positive,

$$-100 = 13 \sin 40^\circ t + \frac{1}{2} (-9.81) t^2 \quad [\text{C1}]$$

$$t = -3.74 \text{ s (rej) or } t = 5.45 \text{ s} \quad [\text{A1}]$$

(ii) $s_x = u_x t = 10(5.45) = 55 \text{ m} \quad [\text{A1}] \text{ allow ecf}$

(iii) $v_y = u_y + a t$

taking upwards as positive,

$$v_y = 13 \sin 40^\circ + (-9.81)(5.45) = -45.1 \text{ m s}^{-1} \quad [\text{C1}]$$

$$v = \sqrt{(45.1)^2 + (10)^2} = 46 \text{ m s}^{-1} \quad [\text{A1}]$$

$$\theta = \tan^{-1}\left(\frac{45.1}{10}\right) = 77^\circ \text{ (clockwise) below horizontal} \quad [\text{A1}]$$

